INCH-POUND

MIL-C-24640/22C
26 June 1995
SUPERSEDING
MIL-C-24640/22B

MILITARY SPECIFICATION SHEET

CABLE, ELECTRIC, 600 VOLTS, LIGHT WEIGHT, SEVEN CONDUCTOR, POWER, TYPE 7XW (INCLUDING VARIATION TYPE 7XWA)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation: MIL-C-24640.

CONSTRUCTION (Watertight)

FIRST - Copper conductor, uncoated in accordance with ASTM B-8, 7 strands (watertight). AWG size and dimensions in accordance with tables I or II for the applicable type.

SECOND - Mica glass tape insulation 0.0029 ±.0002 inch (50 percent overlap maximum and minimum overlap of 45 percent).

THIRD - Opaque-pigmented polyimide tape .001 inch (50 percent overlap minimum).

FOURTH - Opaque-pigmented polyimide tape .001 inch with 0.0001 inch Fluorinated Ethylene Propylene [FEP] (50 percent overlap minimum).

FIFTH - Modified aromatic polyimide resin coating 0.0005 inch minimum (.001 inch nominal). Identification codes applied using Method 1 or 3.

SIXTH - Seven component wires shall be cabled together. Cabling shall be in sequence starting with number 1 from center outward. Fillers may be used as necessary, to form a round cable.

SEVENTH - Binder tape applied helically with overlap.

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CONSTRUCTION VARIANT FOR TYPE 7XW

EIGHTH - Crosslinked polyolefin jacket. (See Table I for thickness).

CONSTRUCTION VARIANT FOR TYPE 7XWA

EIGHTH - Crosslinked polyolefin jacket. (See Table II for thickness).

NINTH - Braided metal armor. Dimensions in accordance with Table II.

TABLE I. Construction details - unarmored.

Pin M24640/20	Type Number	AWG size	Conductor diameter (inch) (nominal)	Component wire diameter (inch) (min)	(max)	Jacket thickness (inch) (minimum)	Cable diameter (inch) (min)	(max)	Cable weight (pounds per 1000 feet) (max)	Conductor resistance per 1000 feet (ohms)
-01UN	7XW-3	16	0.058	0.075	0.081	0.038	0.315	0.339	110.0	4.35
-02UN	7XW-4	14	.073	.091	.097	.041	.374	.404	162.0	2.88

TABLE II. Construction details - armored.

Pin M24640/20	Type Number	AWG size	Conductor diameter (inch) (nominal)	Component wire diameter (inch) (min)	(max)	Jacket thickness (inch) (minimum)	Cable diameter (inch) (min)	(max)	Cable weight (pounds per 1000 feet) (max)	Conductor resistance per 1000 feet (ohms)
-01AN	7XWA-3	16	0.058	0.075	0.081	0.038	0.365	0.389	156.2	4.35
-02AN	7XWA-4	14	.073	.091	.097	.041	.424	.454	229.5	2.88

REQUIREMENTS:

Qualification required.
Construction details: See tables I, II and III.

Component wire:

For polyimide tape constructions, the following additional tests shall be applicable in accordance with MIL-W-81381:

Temperature rating: 150°C (302°F) maximum conductor temperature
Voltage rating: 600 volts (rms) at sea level.
Blocking: Oven temperature, 200+ 2°C (392± 3.6°F) no sticking of conductors
Flammability: 3 seconds (max) after flame.
3.0 inches (max) flame travel.
No flaming of tissue paper.
Humidity resistance: 5 megohms for 1000 feet, minimum insulation resistance after humidity exposure.

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Impulse dielectric test: 100 percent test. Impulse voltage at 8.0 kV.
Insulation resistance: 2500 megohms for 1000 feet (min).
Polyimide cure test: No cracking.
Shrinkage: 0.062 inch maximum at 180°C
Surface resistance: 5 megohms inches (min), initial and final readings.
Thermal shock: Oven temperature, 150± 2°C (302± 3.6°F). Maximum change in measurement: 0.031 inch.
Wet dielectric test: 2500 volts (rms).

EXAMINATION AND TESTS:

Basic electrical:	Requirements
Conductor Resistance - ohms/1000 feet at 25°C, maximum	See tables I and II
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor	2000
Conductor to shield	Not applicable
Shield to shield	Not applicable
Insulation resistance - megohms/1000 feet, minimum	2500
Conductor and shield continuity	No failure
Jacket flaws	Not applicable
Group A:	
Visual and dimensional	No failure
Capacitance:	
Mutual capacitance - at 1 MHz, picofarads/foot (pF/ft), maximum	Not applicable
Capacitance unbalance - percent, maximum	Not applicable
Characteristic impedance - at 1 MHz, ohms	Not applicable
Watertightness (see table XI of MIL-C-24640 for limits of water leakage)	No failure
Weight	No failure
Group B:	
Cold bending	No failure
Drip	No failure
* Cross link proof test (percent maximum)	
Jacket	50
Tear - pounds per inch thickness, minimum	35
Physicals (unaged)	
Jacket	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum	160
Attenuation - at 3 MHz, dB/100 feet, maximum	Not applicable
* Gas flame (1 hour)	No failure

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	Group C:	
	Physicals (aged)	
	Jacket	
	Air oven	
	Tensile strength - percent of unaged, minimum	60
	Elongation - percent of unaged, minimum	60
	Hot oil immersion	
*	Tensile strength - percent of unaged, minimum	50
*	Elongation - percent of unaged minimum	50
*	Permanence of printing (conductor) - method 1 only cycles, minimum	25
*	Permanence of printing (jacket) - cycles, minimum	125
	Armor - conformance to material, (type 7XWA only) construction and coverage	No failure
	Shield - conformance to material, construction and coverage	Not applicable
	Heat distortion - percent of unaged, maximum	30
*	Cable sealant removability	No failure
	Shrinkage	No failure
	Mutual inductance - dB/1000 feet, maximum	Not applicable
	Surface transfer impedance	
	Milliohms per meter, maximum	Not applicable
	EMP response, dB, minimum	Not applicable
	Pulse response time - ms/500 feet	
	Rise time, maximum	Not applicable
	Decay time, maximum	Not applicable
*		
	Group D	
	Flame propagation	No Failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

	Aging and compatibility (cable)	No failure
	Acid gas equivalent - percent, maximum	
	Jacket	2
	Fillers	2
	Insulation	18
	Halogen content - percent, maximum	
	Jacket	0.2
	Fillers	0.2
	Immersion	
	Tensile strength - percent of unaged, minimum	50
	Elongation - percent of unaged minimum	50
*	Smoke index, maximum	
	Jacket	25
	Fillers	45
	Insulation	10

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Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure
Fungus Resistance	No failure
Abrasion resistance	No failure
Gas Flame (3 hour)	No failure
Component wire	No failure

Unit ordering length: 500 feet nominal.

NOTE: The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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